

Runge-Kutta Method

Runge-Kutta methods are designed to give greater accuracy and they possess the advantage of requiring only the function values at some selected points on the subinterval.

$$y_1 = y_0 + \frac{h}{2} [f_0 + f(x_0 + h, y_0 + hf_0)],$$

where $f_0 = f(x_0, y_0)$. If we now set

$$k_1 = hf_0 \quad \text{and} \quad k_2 = hf(x_0 + h, y_0 + k_1)$$

then the above equation becomes

$$y_1 = y_0 + \frac{1}{2} (k_1 + k_2),$$

which is the second-order Runge-Kutta formula. The error in this formula can be shown to be of order h^3 by expanding both sides by Taylor's series.

Higher-order Runge-Kutta formulae exist, of which we mention only the fourth-order formula defined by

$$y_1 = y_0 + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

where

$$\left. \begin{aligned} k_1 &= hf(x_0, y_0) \\ k_2 &= hf\left(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}k_1\right) \\ k_3 &= hf\left(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}k_2\right) \\ k_4 &= hf(x_0 + h, y_0 + k_3) \end{aligned} \right\}$$

in which the error is of order h^5 .

Example: Given $dy/dx = y - x$ where $y(0) = 2$, find $y(0.1)$ and $y(0.2)$ correct to four decimal places.

Solution: (i) Runge-Kutta second-order formula: With $h = 0.1$, we find $k_1 = 0.2$ and $k_2 = 0.21$.

Hence

$$y_1 = y(0.1) = 2 + \frac{1}{2}(0.41) = 2.2050$$

To determine $y_2 = y(0.2)$, we note that $x_0 = 0.1$ and $y_0 = 2.2050$. Hence, $k_1 = 0.1(2.105) = 0.2105$ and $k_2 = 0.1(2.4155 - 0.2) = 0.22155$. It follows that

$$y_2 = 2.2050 + \frac{1}{2}(0.2105 + 0.22155) = 2.4210$$

Proceeding in a similar way, we obtain

$$y_3 = y(0.3) = 2.6492 \quad \text{and} \quad y_4 = y(0.4) = 2.8909$$

(ii) Runge-Kutta fourth-order formula:

To determine $y(0.1)$, we have $x_0 = 0, y_0 = 2$ and $h = 0.1$. We then obtain

$$\begin{aligned} k_1 &= 0.2 \\ k_2 &= 0.205 \\ k_3 &= 0.20525 \\ k_4 &= 0.21053. \end{aligned}$$

Hence

$$y(0.1) = 2 + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) = 2.2052$$

Proceeding similarly, we obtain $y(0.2) = 2.4214$.

Example: Given $dy/dx = 1 + y^2$, where $y = 0$ when $x = 0$, find $y(0.2)$ and $y(0.4)$.

Solution: We take $h = 0.2$. With $x_0 = y_0 = 0$, Then we obtain

$$\begin{aligned} k_1 &= 0.2 \\ k_2 &= 0.2(1.01) = 0.202 \\ k_3 &= 0.2(1 + 0.010201) = 0.20204 \\ k_4 &= 0.2(1 + 0.040820) = 0.20816 \end{aligned}$$

and

$$y(0.2) = 0 + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) = 0.2027$$

which is correct to four decimal places. To compute $y(0.4)$, we take $x_0 = 0.2, y_0 = 0.2027$ and $h = 0.2$

$$\begin{aligned} k_1 &= 0.2 \left[1 + (0.2027)^2 \right] = 0.208 \\ k_2 &= 0.2 \left[1 + (0.3068)^2 \right] = 0.2188 \\ k_3 &= 0.2 \left[1 + (0.3121)^2 \right] = 0.2195 \\ k_4 &= 0.2 \left[1 + (0.4222)^2 \right] = 0.2356 \end{aligned}$$

And

$$y(0.4) = 0.2027 + 0.2201 = 0.4228,$$

correct to four decimal places.